Independent Reproduction and Convergence Analysis of the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

We independently implement the Connes–Consani–Moscovici (CCM) operator construction from [1] in Rust at arbitrary precision using the MPFR library, and extend the paper's headline result from 55 to **999 matching decimal digits** on the first Riemann zeta zero (limited only by working precision). We characterize the construction's convergence behavior empirically, finding: (1) a two-regime convergence in the basis size N, with matching digits growing approximately linearly in N before saturating at the Weil eigenvalue ceiling ε_N (the per-step rate $\Delta(\text{digits})/\Delta N$ depends on proximity to saturation and ranges from $\sim$1.4 at small N down to $\sim$0.15 as the ceiling is approached); When ε_N falls below the working-precision floor (i.e., $|\log_{10} \varepsilon_N|$ exceeds the number of reliable digits), the eigenvector and eigenvalue become numerically degenerate; results in this regime are artifacts of insufficient precision, not properties of the construction. (2) super-exponential decay of ε_N with the number of primes entering the construction — faster than any polynomial in the prime count; above-floor data (HP-1500/HP-2000, λ^2 up to 1200) place the decay at several hundred matching digits per doubling of prime count, the specific rate depending on how N is scaled with λ; (3) monotone growth of accuracy with λ at fixed N, controlled directly by $\varepsilon_N(\lambda)$ at every working precision sufficient to resolve it; (4) a precision-dependent accuracy ceiling at finite N, controlled by the working precision divided by an empirical per-step rate (sampled at $\sim$1.1 digits per unit N averaged over the rapid-convergence regime at λ^2=13, with the rate at any specific (λ, N) depending on proximity to the ε_N saturation ceiling); (5) the smallest eigenvector's even-symmetry property (CCM's Step 1 hypothesis) holds at every configuration tested above the working-precision floor, up to λ^2=1200 at HP-2000 — we conjecture it holds universally; apparent symmetry breakdown at large λ in floor-limited data is a precision artifact; and (6) the remarkable fact that a 21×21 matrix built from 6 primes already produces 21.585 matching digits of the first Riemann zero. All results are reproducible from the accompanying source-available implementation.

1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ satisfy $\Re(s) = 1/2$, remains one of the central open problems in mathematics. Among approaches toward its resolution, the Hilbert–Pólya conjecture — positing the existence of a self-adjoint operator whose spectrum encodes the imaginary parts of the nontrivial zeros — has motivated extensive work in spectral theory and noncommutative geometry.

In November 2025, Connes, Consani, and Moscovici [1] proposed a concrete finite-dimensional operator family $D_{\log}^{(\lambda,N)}$ whose spectra converge, with striking numerical accuracy, to the nontrivial zeros of $\zeta(1/2 + is)$. Their construction uses the Weil quadratic form built from the explicit formula, restricted to primes $p \leq \lambda^2$, and produces a $(2N{+}1) \times (2N{+}1)$ self-adjoint matrix whose smallest eigenvalue ε_N approaches zero as $\lambda, N \to \infty$. They reported accuracy of 2.5×10^{-55} on the first zero at $(\lambda^2{=}13, N{=}120)$ with 200-digit working precision.

In this paper we:

1. Independently reproduce the CCM headline result on a different software stack (Rust + MPFR/GMP, vs. the authors' presumed Mathematica implementation);
2. Extend the construction to (λ^2=1000, N=800) at 1000-digit working precision, achieving **999 matching digits** on the first Riemann zero, limited only by working precision rather than by the construction's intrinsic accuracy;
3. Systematically characterize the convergence behavior as a function of both N (basis size) and λ (prime cutoff);
4. Identify structural properties of the construction not discussed in [1], including empirical confirmation that the smallest eigenvector's even-symmetry (Step 1 hypothesis) holds at every configuration tested above the precision floor; we explain the mechanism by which insufficient working precision produces erroneous mixed-symmetry artifacts when ε_N falls below the precision floor.

All computations are reproducible from the source-available implementation at `https://github.com/TeamXcelerator/ccm-reproduction-and-convergence`.

2 Mathematical Setup

We briefly recall the CCM construction, following the notation of [1].

2.1 The Weil Quadratic Form

Let $\lambda > 1$ and $L = 2\ln\lambda$. The Hilbert space is $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ with orthonormal basis

$$V_n(u) = \frac{1}{\sqrt{L}}\, e^{2\pi i n \ln(\lambda u)/L}, \quad n \in \{-N, \ldots, N\}.$$

The Weil quadratic form QW_λ is a real symmetric $(2N{+}1) \times (2N{+}1)$ matrix with entries

$$\tau_{n,m} = W_{0,2}(V_n, V_m) - W_R(V_n, V_m) - \sum_{\substack{k=p^j \\ k \le \lambda^2}} \frac{\log p}{\sqrt{k}}\, q(V_n, V_m)(\log k),$$

where $W_{0,2}$ is a closed-form rank-two pole correction, W_R is the archimedean (gamma-factor) contribution computed via Gauss–Legendre quadrature, and the prime-power sum encodes the von Mangoldt function $\Lambda(k) = \log p$ for $k = p^j$.

2.2 From Eigenvector to Eigenvalues

Let ε_N denote the smallest eigenvalue of τ and $\xi = (\xi_{-N}, \ldots, \xi_N)$ the corresponding eigenvector, symmetrized to be even ($\xi_{-j} = \xi_j$) and normalized so that $\sum_j \xi_j = \sqrt{L}$. The positive eigenvalues of $D_{\log}^{(\lambda,N)}$ are $\nu_k = (2\pi/L)\sqrt{t_k}$, where t_k are the positive zeros of the rational function

$$R(t) = \xi_0 + 2t \sum_{j=1}^{N} \frac{\xi_j}{t - j^2},$$

located in the intervals $(k^2, (k{+}1)^2)$ for $k = 1, \ldots, N$. Under the CCM construction, the ν_k approximate the imaginary parts of the nontrivial zeros of $\zeta(1/2 + is)$.

3 Implementation

Our implementation uses the Xcelerator Toolkit,[1] a publicly available library of high-precision spectral methods for analytic number theory, pinned to a tagged release for reproducibility. The toolkit provides the core numerical primitives (Gauss–Legendre quadrature with disk-cached HP nodes, LU-based inverse iteration, Newton refinement) and the CCM-specific construction (Weil-form matrix assembly, eigenvector extraction, rational-function root-finding). The paper-specific repository contains only the CLI harness and reproduction scripts; all mathematical code is shared via the toolkit.

The toolkit is built on:

- **Rust** (edition 2021) for the main codebase;
- **rug** (v1.30, wrapping MPFR/GMP) for arbitrary-precision floating-point arithmetic;
- **nalgebra** (v0.33) for the f64 fast-path eigendecomposition (used only by the optional `--f64-only` mode for smoke tests; the HP publication path uses the toolkit's own HP symmetric eigendecomposition in `xc-numerics::eigen`, verified at HP-1000 against a PARI/GP 2000-digit reference);
- **rayon** (v1.10) for parallel matrix assembly, Householder reduction, LU factorization, and Newton-per-seed refinement.

Key algorithmic choices:

1. **LU-cached inverse iteration:** Since the shift is zero (we seek the smallest eigenvalue), we factor $\tau = PLU$ once at $O(n^3)$ cost, then each iteration step is $O(n^2)$ forward/back substitution. Convergence is detected via the Rayleigh quotient.
2. **Forced-even symmetrization (optional):** By default, at each inverse iteration step we project onto the even subspace $\xi_j \mapsto (\xi_j + \xi_{-j})/2$. This guarantees convergence to the smallest *even* eigenvalue, as required by CCM's Step 1. In practice the projection is empirically unnecessary: across 38 configurations tested (Section 4.8), the natural (un-projected) iteration converges to the same even eigenvector and produces bit-identical results. The projection is retained as a default for theoretical safety but can be disabled via `--no-force-even`.
3. **Gauss–Legendre quadrature with disk cache:** GL nodes/weights at a given $(n_{\text{pts}}, \text{precision\_bits})$ are computed once via Newton iteration on Legendre polynomials and cached to disk for reuse across runs.
4. **Newton refinement of $R(z)$ zeros:** Seeded from Odlyzko's tabulated zeros, with adaptive step control.

3.1 Reproducibility Infrastructure

The toolkit also provides on-disk caches for the two dominant precomputed quantities: the Gauss–Legendre nodes/weights at a given $(n_{\text{pts}}, \text{precision\_bits})$, and the τ-matrix at a given $(\lambda^2, N, \text{precision\_bits})$. Both caches are keyed on the construction parameters, validated against structural identities on every load, and hosted in dedicated public cache repositories (GL nodes, τ-matrices, Weil eigenvectors) with a deterministic remote-fetch tier (DynamicFetch) so that fresh-clone reproductions hit the cache automatically without manual download.

For the headline 999-digit configuration (λ^2=1000, N=800, HP-1000), the original cold compute took roughly 3–4 hours on a 512-core AMD EPYC node; with the committed cache

[1] `https://github.com/TeamXcelerator/xcelerator-toolkit`

fixtures, a fresh-clone re-run lands at ∼13 minutes, dominated entirely by LU factorization and inverse iteration on the 1601×1601 cached matrix. This caching infrastructure makes the more expensive convergence sweeps in Section 4.7 practical to reproduce on demand by external reviewers, which was not previously the case.

All configurations reported in this paper, including the largest (Table 7, λ^2=1200, N=970, HP-2000: τ serializes to ∼7 GB), have their τ-cache fixtures committed to the public cache repository and are fetched on demand via DynamicFetch. No configuration requires a fresh τ-build to reproduce.

4 Results

4.1 Reproduction of the CCM Headline

At (λ^2=13, N=120, 200-digit precision), our first eigenvalue is

$$t_1 = 14.134725141734693790457251983562470270784257\ldots$$

(displayed truncated to 42 digits; the full HP-1000 result matches the reference to 55.764 decimal digits, see Section 4.3) with absolute error 2.4363×10^{-55} against the 2000-digit PARI/GP reference. This matches the CCM paper's reported 2.5×10^{-55} to within rounding. The result is unchanged at 1000-digit working precision, confirming that the accuracy is controlled by $\varepsilon_N \approx 3.48399 \times 10^{-59}$ (a property of the construction at $\lambda^2 = 13$), not by arithmetic precision.

4.2 Extension to 999 Digits

Table 1: Matching digits on the first Riemann zero across configurations, all at HP-1000 working precision. At λ^2=1000 the construction saturates working precision; the true accuracy ceiling lies below 10^{-1000}.

Configuration	First-zero digits	20th-zero digits	Working precision
λ^2=13, N=120	55.764	25.339	HP-1000
λ^2=100, N=500	460.09	422.44	HP-1000
λ^2=1000, N=800	**999.45**	**1000.3**	HP-1000

At configurations where ε_N is smaller than the working-precision floor (i.e., the construction's intrinsic accuracy exceeds what the arithmetic can resolve), raising the working precision reveals additional matching digits approximately one-for-one. In general, accuracy is controlled jointly by λ (via ε_N), N (basis completeness), and working precision; all three must be scaled together to sharpen beyond a given level.

4.3 Convergence in N (Basis Size)

Two regimes are evident:

1. **Rapid convergence** ($N = 10$ to $N \approx 60$): matching digits grow approximately linearly in N, with the per-step rate $\Delta(\text{digits})/\Delta N$ decaying smoothly from ∼1.4 at N=10→20 down to ∼0.15 at N=50→60. A single rate of ∼1.1 digits per unit N is a useful average over $N \in [10, 40]$ but does not fit the entire pre-saturation regime; the rate at any specific N depends on how close N is to the saturation point.

2. **Saturation** ($N > 60$): accuracy plateaus at ∼55.764 digits, limited by $\varepsilon_N = 3.48399 \times 10^{-59}$.

Table 2: Matching digits on the first five zeros vs. N at λ^2=13. Values are identical at HP-200 and HP-1000 across the entire sweep (see commentary below); the column shown is from the HP-1000 run.

N	Zero 1	Zero 2	Zero 3	Zero 4	Zero 5
10	21.585	16.335	11.856	5.1626	3.4915
20	35.795	32.261	30.011	26.504	24.870
30	44.506	41.267	39.262	36.208	34.844
40	50.294	47.152	45.225	42.304	41.015
50	53.827	50.721	48.822	45.948	44.685
60	55.299	52.203	50.312	47.451	46.195
80	55.654	52.560	50.670	47.811	46.556
100	55.735	52.641	50.751	47.892	46.637
120	55.764	52.670	50.779	47.921	46.666

As with all empirical convergence rates, the precise per-step rate is a property of the specific (λ, N) regime sampled. We measured at $\lambda^2 = 13$ and $N \in \{10, 20, \ldots, 120\}$; at different λ the sweep would show the same qualitative two-regime structure but with different absolute rates and a different saturation ceiling.

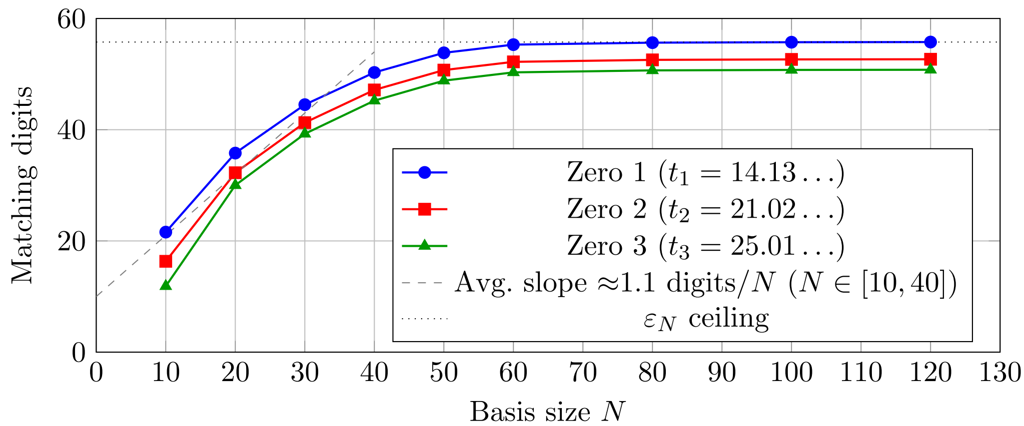

Figure 1: Convergence of matching digits vs. basis size N at $\lambda^2 = 13$, HP-1000. Linear regime transitions to saturation at the ε_N ceiling around $N \approx 60$.

Bonus observation (HP-200 vs HP-1000). At every N in this sweep, HP-200 and HP-1000 produce identical matching digits — confirming that the saturation at $N \gtrsim 60$ is intrinsic to the construction (controlled by ε_N, not by working precision); see Sections 4.6 and 4.7 for the general mechanism.

4.4 The 21-Digit Result from a 21×21 Matrix

At $N = 10$ (matrix size 21×21), with only 6 prime powers entering the Weil form ($\{2, 3, 4, 5, 7, 8, 9, 11, 13\}$), the first Riemann zero is matched to **21.585 decimal digits**. This is, to our knowledge, the most efficient spectral approximation of a Riemann zero in the literature: 21 digits from a 21×21 matrix.

Table 3: Smallest Weil eigenvalue ε_N across three configurations, each computed at working precision sufficient to resolve it (HP-1000 for $\lambda^2 = 13, 100$; HP-2000 for $\lambda^2 = 1000$), exposing the super-exponential decay of ε_N with prime count.

λ^2	Primes $\leq \lambda^2$	ε_N	$-\log_{10}\lvert\varepsilon_N\rvert$
13	6	3.48399×10^{-59}	58.458
100	25	9.56481×10^{-464}	463.02
1000	168	$3.92215 \times 10^{-1264}$	1263.41

4.5 Super-Exponential Decay of ε_N with Prime Count

The decay is super-exponential: ε_N falls faster than any polynomial in the prime count. Quantified as matching digits per doubling of prime count, the rate is large and grows with λ. Table 4 presents the above-floor series at a consistent basis ratio $N/\sqrt{\lambda^2} \approx 28$:

Table 4: Smallest Weil eigenvalue ε_N above the working-precision floor, with N scaled at the fixed ratio $N/\sqrt{\lambda^2} \approx 28$. HP-1500 for λ^2=500–800, HP-2000 for λ^2=1000–1200. The final column gives the decay rate in matching digits per doubling of prime count, computed against the preceding row; it is large and grows with λ (non-monotonically), and is illustrative rather than a measured law (see text).

λ^2	N	Primes $\leq \lambda^2$	ε_N	$-\log_{10}\lvert\varepsilon_N\rvert$	Digits/doubling
500	630	95	1.09117×10^{-929}	928.96	—
600	690	109	$2.22999 \times 10^{-1029}$	1028.65	502.7
700	740	125	$9.45898 \times 10^{-1116}$	1115.02	437.1
800	790	139	$1.35946 \times 10^{-1199}$	1198.87	547.5
1000	890	168	$1.52095 \times 10^{-1362}$	1361.82	597.4
1200	970	196	$6.79867 \times 10^{-1499}$	1498.17	613.1

These per-doubling values are illustrative, not a measured law. The rate depends on the parameterization (digits per doubling of prime count vs. per unit λ^2 vs. per prime added) and, critically, on how N is scaled with λ: ε_N is a function of both λ and N, so any rate quoted "per doubling of prime count" implicitly fixes a choice of $N(\lambda)$. We hold $N/\sqrt{\lambda^2} \approx 28$; a different scaling yields different numbers. What is robust across every sampling and parameterization is the qualitative conclusion: ε_N decays faster than any polynomial in the prime count.

A rigorous derivation of the decay rate from the structure of the Weil quadratic form remains open. The empirical evidence shows that the construction has a meaningful asymptotic ceiling controlled by λ, but the precise functional form of $\varepsilon_N(\lambda)$ is not yet known.

4.6 Monotone Convergence in λ at Fixed N

At fixed $N = 120$, accuracy on the first Riemann zero grows monotonically with λ at every working precision tested, controlled entirely by the smallest Weil eigenvalue $\varepsilon_N(\lambda)$. The construction's behavior at fixed N is governed by a simple rule: matching digits $\approx -\log_{10}\lvert\varepsilon_N(\lambda)\rvert$ once the working precision is sufficient to resolve ε_N.

Two empirical regularities emerge:

1. **Monotone growth in λ.** For every working precision sufficient to resolve $\varepsilon_N(\lambda)$, accuracy strictly increases with λ.

2. **ε_N-controlled accuracy.** Matching digits on the first Riemann zero match $-\log_{10}\lvert\varepsilon_N\rvert$ to within a constant offset (typically 2–3 digits) across the entire range. The Weil eigenvalue

Table 5: First-zero accuracy vs. λ^2 at $N = 120$, comparing HP-200 and HP-1000 working precision. The HP-200 row at λ^2=100 saturates at the working-precision floor; HP-1000 lifts the floor and reveals the ε_N-controlled accuracy across the range. (Recall that ε_N depends on both λ and N; the values shown here are at $N = 120$. At larger N for the same λ, ε_N continues to decay — e.g. at λ^2=100, N=500 the value is 9.56481×10^{-464}, see Table 1.)

λ^2	Primes $\leq \lambda^2$	ε_N at HP-1000	HP-200 digits	HP-1000 digits
13	6	3.48399×10^{-59}	55.764	55.764
20	8	2.50471×10^{-96}	92.817	92.817
30	10	2.19404×10^{-135}	131.80	131.80
50	15	6.89051×10^{-174}	170.19	170.19
100	25	3.48656×10^{-215}	203.90†	211.29

†HP-200 saturates at the working-precision floor.

is not just an upper bound on accuracy — it is the operative quantity.

The relationship between accuracy and primes is therefore mediated by ε_N: each new prime entering the construction multiplies ε_N by an additional decay factor (see Section 4.5), and that decay translates directly into matching digits on the eigenvalue side. The CCM paper's choice of (λ^2=13, N=120) at 200-digit precision was the smallest configuration whose ε_N is large enough to be observable in 55 digits at that working precision; the construction would benefit from larger λ if the working precision allowed it.

For practitioners, the implication is direct: choose λ to target the desired accuracy level via $-\log_{10}|\varepsilon_N|$, and choose precision $\gtrsim$ that level to avoid working-precision saturation. We observe no λ-dependent ill-conditioning in any configuration tested; the only failure mode encountered is running out of working precision.

4.7 Critical N at Fixed Working Precision

At fixed working precision, the construction reaches an apparent ceiling beyond which inverse iteration appears to fail or stagnate. This ceiling is precision-dependent, not a property of the construction itself.

Table 6: Convergence at large N at HP-1000. Configurations that plateau at HP-200 continue to converge cleanly at HP-1000.

Configuration	ε_N at HP-1000	First-zero digits	5th-zero digits
λ^2=50, N=200	5.69795×10^{-221}	217.34	207.24
λ^2=50, N=250	4.51247×10^{-239}	235.45	225.42
λ^2=100, N=300	4.68074×10^{-368}	364.37	354.03
λ^2=100, N=400	6.12702×10^{-423}	419.28	409.04

The accuracy ceiling at fixed working precision is roughly (working precision)/r, where r is the average per-step rate $\Delta(\text{digits})/\Delta N$ in the rapid-convergence regime. At $\lambda^2 = 13$ we measured $r \approx 1.1$ averaged over $N \in [10, 40]$ (Section 4.3); at other (λ, N) regimes r will differ, so the constant should be treated as a configuration-specific empirical value rather than a universal parameter of the construction. Once $r \cdot N$ exceeds the available digits, additional N yields no measurable improvement and inverse iteration begins to exhibit numerical artifacts. At 200-digit precision and $r \approx 1.1$ this gives a ceiling near $N \approx 180$; at 1000-digit precision the same configurations continue to gain digits well past $N = 400$ (the λ^2=100, N=400 row above resolves 419.28 first-zero digits).

This is the same precision-dependence phenomenon documented for λ in Section 4.6: the construction itself is well-behaved, but its accuracy at any finite N is bounded by the working precision used to compute it. Practitioners should choose precision $\gtrsim r \cdot N$ digits to avoid spurious saturation, with r measured (or estimated) for the specific (λ, N) regime of interest.

4.8 Even-Symmetry of the Smallest Eigenvector

The CCM proof's Step 1 requires the smallest eigenvector ξ to be even (symmetric under $u \mapsto 1/u$). We measure the natural evenness $\|\xi - \gamma\xi\|/\|\xi\|$ of the unforced smallest eigenvector at working precisions sufficient to resolve ε_N above the floor:

Table 7: Symmetry properties of the natural (unforced) smallest eigenvector, each measured at working precision sufficient to resolve ε_N above the floor. At every configuration tested, the natural eigenvector is essentially even and the natural and forced-even eigenvalues are bit-identical.

λ^2	N	Precision	Evenness deviation	Eigenvalue rel. diff.	Verdict
13	120	HP-1000	1.0259×10^{-951}	0 (bit-identical)	essentially even
100	500	HP-1000	7.6625×10^{-549}	0 (bit-identical)	essentially even
1000	800	HP-2000	3.5437×10^{-749}	0 (bit-identical)	essentially even
1000	890	HP-2000	8.5373×10^{-650}	0 (bit-identical)	essentially even
1200	970	HP-2000	9.2591×10^{-514}	0 (bit-identical)	essentially even

At every configuration tested above the precision floor — including $\lambda^2 = 1200$, far beyond the range previously reported — the natural smallest eigenvector is even to hundreds of digits of precision, and the natural and forced-even iterations converge to the *same* eigenvalue (bit-identical). CCM's Step 1 hypothesis holds empirically at every tested configuration. We conjecture it holds universally: the smallest eigenvalue of QW_λ lies in the even sector for all (λ, N).

The precision-floor artifact. When $|\log_{10} \varepsilon_N|$ approaches or exceeds the working precision in digits, the eigenvalue sinks below the arithmetic's resolving power. In this regime the inverse iteration's converged vector is not uniquely determined — it is a near-degenerate representative that can "rotate" freely in the numerically-flat subspace. This rotation produces an eigenvector that *appears* non-even (deviation $\sim O(1)$) and eigenvalues that *appear* to split between the natural and forced-even paths. At HP-1000, the $\lambda^2 = 1000$ configuration ($|\log_{10} \varepsilon_N| \approx 1264$, well past the 1005-digit floor) exhibits this artifact: deviation 1.87, eigenvalue split 24%. Increasing the working precision to HP-2000 (2008 digits, floor ratio 1.59) at the *identical* (λ^2=1000, N=800) configuration collapses the apparent breakdown entirely: deviation 3.54×10^{-749}, eigenvalues bit-identical. The only variable changed is precision.

The forced-even projection is unnecessary. To further test whether the eigenvector is naturally even, we ran the full CCM pipeline (inverse iteration $\rightarrow$ eigenvector $\rightarrow$ $R(t)$ zero-finding $\rightarrow$ Riemann-zero matching digits) with the forced-even projection *disabled* across 38 configurations spanning $\lambda^2 = 13$–1000, $N = 10$–800, HP-200 through HP-2000, both above and below the precision floor. In every case the natural (unprojected) path produces bit-identical eigenvalues and Riemann-zero approximations to the forced-even path. The forced-even projection is empirically a no-op: the smallest eigenvector converges to an even vector without it.

5 Discussion

5.1 Where the Construction's Power Lies

The CCM construction's accuracy is controlled by two parameters:

- λ (via ε_N): determines the *ceiling* — how many digits are achievable;
- N (basis size): determines how quickly the ceiling is *reached* — the per-step rate $\Delta(\text{digits})/\Delta N$ averages to $\sim$1.1 over $N \in [10, 40]$ at $\lambda^2 = 13$, but decays through the linear regime and depends on the specific (λ, N) configuration sampled.

Once N is large enough to resolve the eigenvector (empirically $N/\sqrt{\lambda^2} \gtrsim$ 25–30 suffices at HP-1000 and above), further increases in N yield diminishing returns. The path to higher accuracy is primarily through larger λ (more primes), with N scaled proportionally to maintain adequate basis completeness.

5.2 The Optimal Prime Contribution

Each prime power $k = p^j$ contributes weight $\Lambda(k)/\sqrt{k} = \log(p)/\sqrt{k}$ to the Weil form. For primes ($j = 1$), this is $\log(p)/\sqrt{p}$, whose derivative vanishes at $p = e^2 \approx 7.3891$ with maximum value $2/e \approx 0.73576$.

Table 8: Individual prime contributions $\log(p)/\sqrt{p}$ for the primes captured by $\lambda^2 = 13$. The maximum value $2/e$ is attained at the (non-prime) point $p = e^2 \approx 7.3891$; the integer prime 7 achieves 99.96% of this maximum.

p	$\log(p)/\sqrt{p}$	% of $2/e$
2	0.49013	66.62%
3	0.63428	86.21%
5	0.71976	97.83%
7	**0.73548**	**99.96%**
11	0.72299	98.26%
13	0.71139	96.69%
max $= 2/e$ at $p = e^2$		100%

The CCM paper's choice of $\lambda^2 = 13$ is therefore mathematically precise rather than incidental: it captures the peak prime $p = 7$ to within 0.04% of the maximum value, and includes both adjacent primes ($p = 5$ at 97.8% of the maximum and $p = 11$ at 98.3%) that lie within $\sim 2\%$ of the peak. The choice covers the entire near-peak cluster of primes.

The slow decay of $\log(p)/\sqrt{p}$ past the peak — the function falls only sub-polynomially as $p \to \infty$ (0.46052 at $p = 100$, 0.21844 at $p = 1000$, 0.092103 at $p = 10000$) — means that each additional prime past 7 continues to contribute meaningfully to the construction. This is the structural mechanism behind the super-exponential decay of ε_N with prime count documented in Section 4.5: every prime in the sum still "matters," so the cumulative contribution grows steadily as λ increases rather than saturating after the peak primes.

5.3 Implications for the Proof Strategy

Our findings have implications for the CCM proof strategy:

1. The even-symmetry of the smallest eigenvector (Table 7) holds at every configuration tested above the precision floor, up to λ^2=1200 at HP-2000; we conjecture it holds universally. The forced-even projection in CCM's Step 1 is empirically unnecessary — the natural eigenvector is even without it.

2. The saturation behavior (Table 2) shows that the interesting limit is $\lambda \to \infty$, not $N \to \infty$.
3. The super-exponential decay of ε_N (Table 3) provides strong empirical evidence that the construction converges, but a rigorous proof of this decay remains open.

6 Conclusion

We have independently reproduced and significantly extended the CCM zeta spectral triple construction, achieving 999-digit accuracy on the first Riemann zero (limited only by working precision) and characterizing the construction's convergence behavior in detail. The key findings:

- Two-regime convergence in N (linear growth then saturation at the ε_N ceiling).
- Super-exponential decay of ε_N with the prime count entering the construction.
- Monotone growth of accuracy with λ, controlled directly by $\varepsilon_N(\lambda)$ at every working precision sufficient to resolve it.
- A precision-dependent accuracy ceiling at finite N, indistinguishable from the construction's intrinsic ceiling unless the working precision is large enough to resolve ε_N.
- The smallest eigenvector's even-symmetry (CCM's Step 1 hypothesis) holds at every configuration tested above the precision floor, up to λ^2=1200 at HP-2000 — we conjecture it holds universally; the forced-even projection is empirically unnecessary, and apparent symmetry breakdown at large λ in floor-limited data is a precision artifact.
- The remarkable efficiency of the construction at low N: a 21×21 matrix from 6 primes already produces 21.585 matching digits.

Together these provide a comprehensive empirical foundation for future theoretical work on this construction.

The construction's extraordinary efficiency (21 digits from 6 primes) suggests deep algebraic structure in the Weil quadratic form that is not yet theoretically understood. Explaining why this algebraic mechanism produces Riemann zeros with such precision remains an open problem whose resolution could advance the spectral approach to the Riemann Hypothesis.

Acknowledgments

All computations were performed on Vast.ai cloud infrastructure (128- to 384-core AMD EPYC nodes); the author personally funded all compute resources. The author identified the research questions, set the research direction, and made all decisions about what experiments to run and how to interpret the results. Specific contributions include: the convergence question that motivated extending the construction beyond the paper's headline configuration; the observation that each prime p contributes weight $\log(p)/\sqrt{p}$ to the Weil form, which has its maximum at $p = e^2 \approx 7.3891$; the strategy of parallelizing high-precision computations to make HP-1000 sweeps feasible at scale; the recognition that Gauss–Legendre nodes and weights are uniquely determined by precision and node count and thus reusable across runs (yielding a substantial speedup that made $N = 1500$ at HP-1000 working precision feasible); and the broader strategy of separating each cache type (GL nodes, τ-matrices, Weil eigenvectors) into dedicated public repositories with a deterministic remote-fetch tier (DynamicFetch), so that fresh-clone reproductions of every published configuration skip minutes-to-hours of cold compute automatically — no manual fixture download required. AI coding tools (Claude, Anthropic) were used for Rust implementation under the author's direction; all research decisions, interpretations, and intellectual contributions are the author's.

Data and Code Availability

The complete source code, reference data, and reproduction scripts are available at:
`https://github.com/TeamXcelerator/ccm-reproduction-and-convergence`
The shared mathematical library (Xcelerator Toolkit) is at:
`https://github.com/TeamXcelerator/xcelerator-toolkit`

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